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Low-temperature fabrication of plasmonic titanium nitride thin films by electron beam evaporation

Anindita Das[1], **Yoshiko Nanao**[1], **Akhil Rajan**[1], **Andrea Di Falco**[1] and **Sebastian A Schulz**[1,2,*]

[1] SUPA, School of Physics and Astronomy, University of St Andrews, North Haugh St Andrews KY16 9SS, United Kingdom
[2] Department of Electrical and Computer Engineering, Waterloo Institute of Nanotechnology, University of Waterloo, 200 University Avenue W, Waterloo N2L 3G1 ON, Canada
[*] Author to whom any correspondence should be addressed.

E-mail: sas35@st-andrews.ac.uk and sebastian.schulz@uwaterloo.ca

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Abstract

Titanium nitride (TiN) is a refractory metal nitride compound with the potential to function as an alternative plasmonic material to gold, especially for high-power linear and nonlinear applications such as materials processing, telecommunications, and laser systems. This paper presents a simple, electron beam evaporation based deposition process to fabricate sub-100 nm TiN thin films at low temperatures. The dependence of optical properties and film thickness on deposition temperature was explored and it was observed that low-temperature annealing during deposition further enhances the plasmonic properties of the films. These results are essential for promoting the utilisation of highly metallic TiN thin films in plasmonic and photonic applications.

1. Introduction

Plasmonic materials have a variety of applications ranging from biochemical sensing [1, 2], subdiffraction waveguiding, optical modulators, nonlinear optics, nanolasers development and optical metasurfaces [3–5]. The predominant plasmonic materials are elemental gold and silver due to their excellent conductivity, low optical loss, and biocompatibility. Despite the extensive applications of these metals, some major drawbacks restrict their further use. Noble metals are not abundant and hence costly, constraining their extensive use. The low melting temperature and plasmonic local heating of nanoscale gold and silver cause thermal expansion and shape distortion, limiting their use in high-power linear and nonlinear applications [6, 7]. Further, their incompatibility with typical complementary metal-oxide-semiconductor (CMOS) fabrication techniques is another problem as gold introduces acceptor and donor levels within silicon's intrinsic band gap, leading to deep-level traps that cause charge carrier recombination, material contamination and loss of current [8]. This severely limits the integration of plasmonic or metamaterial components with nanoelectronic devices. To address these fundamental challenges, transition metal nitrides (TMNs), particularly titanium nitride (TiN), have attracted significant interest as alternative plasmonic materials for photothermal applications [9], hot electron generation [10], and photonic integration technologies [11].

In general, TMNs are nonstoichiometric, conductive compounds with large free-carrier concentrations. In particular, TiN, a Group IVB metal nitride, is highly refractory, hard, stable, and naturally abundant. It offers optical tunability in the visible and near-infrared (NIR) wavelengths by varying the metal-to-nitrogen stoichiometry and controlling oxygen impurities. With decomposition temperatures up to 2900 °C and optical properties stable up to 1000 °C, TiN offers exceptional thermal resilience [7, 12]. Further, its gold-like hue, a real part of the permittivity similar to gold but with a higher imaginary part, mechanical durability, cost-effectiveness, and compatibility with silicon CMOS technology make it a promising and suitable alternative to gold. As broad comparisons between TiN and conventional plasmonic materials (e.g. gold, silver) have been well-documented in existing reviews [7, 13]. However, extremely high deposition temperatures are needed to optimize TiN thin-film optical properties. Deposition schemes such as pulsed laser deposition [14], chemical vapour deposition [15], atomic layer deposition [16] and magnetron sputtering [17, 18] have been commonly explored. High-temperature epitaxy (>400 °C) on a crystallographically matched substrate is typically required to achieve films with optimal plasmonic properties. However, such high-temperature processing presents significant challenges as they could lead to doping migration, annealing, or polymer reflow when used in complex, multilayered devices. This issue is a focus of current research in the plasmonic field, aiming to develop CMOS-compatible low-temperature deposition techniques, particularly around room temperature, for producing high-quality TiN thin films [19–21]. Low-temperature deposition techniques for TiN enable precise control over stoichiometry, carrier concentration, and grain structure, significantly influencing its plasmonic properties and stability. A low temperature-processed TiN maintains a high density of plasmonically active electrons, facilitating efficient hot carrier generation crucial for photocatalysis and solar energy harvesting. Additionally, the low processing temperature is critical for combining TiN with temperature-sensitive materials like transparent conductive oxides and flexible substrates. Controlling ENZ (epsilon-near-zero) properties provide precise control of plasmonic performances across visible to infra-red, enhancing applications in imaging, biosensing, and nanophotonics [22, 23]. In recent years, low-temperature magnetron sputtering techniques for depositing highly plasmonic TiN on various substrates under diverse deposition conditions have been reported. Chang *et al* reported a TiN film deposition via room temperature, low-power, bias-free reactive sputtering technique [21]. Most recently, Nieborek *et al* strategically optimized the plasmonic properties of relatively thick, non-transparent polycrystalline TiN films, which were deposited at room temperature using the pulsed-DC reactive magnetron sputtering technique on a non-matched substrate [24]. Although the reported protocols for room-temperature deposition of TiN films with excellent plasmonic properties through sputtering have been successful, they require precise control of various process parameters, such as RF power and reactive gas flow ratio.

Electron beam (e-beam) evaporation offers distinct advantages over other thin-film deposition methods, including high deposition rates, the ability to evaporate high-melting-point materials, and low contamination levels, making it a straightforward and efficient process. Further, e-beam evaporation is a highly directed deposition method, rendering it particularly appropriate for the fabrication of plasmonic nanostructures using lift-off procedures. Despite its promising potential, a comprehensive protocol for growing TiN films using e-beam evaporation has not been thoroughly investigated to date. This study involves the deposition of plasmonic TiN films using the e-beam evaporation technique. It further aims to investigate the optical and structural properties of sub-100 nm TiN thin films, focusing on those annealed at low temperatures, which exhibit strong plasmonic behaviour. By targeting low-temperature deposition, we ensure its compatibility with other sensitive fabrication materials and potential applications in CMOS-compatible photonic and optoelectronic devices. We achieved excellent plasmonic properties in terms of permittivity, comparable to those of polycrystalline gold and silver films, at deposition temperatures slightly above room temperature, in the range of 40 °C–50 °C. Our evaporated TiN films have significant advantages, such as low-temperature processing, directional deposition, and enhanced crystallinity and smoothness, excelling previously reported low-temperature sputtering methods. A comprehensive optical, structural, and morphological analysis of the fabricated films is also presented.

2. Fabrication and characterization techniques

To optimize the plasmonic properties of TiN thin films, we employed the e-beam evaporation technique using the Amod PVD platform from Angstrom Engineering using TiN pellets as the source material for the deposition. All TiN films were deposited on AmScope microscope glass slides, and all substrates were thoroughly cleaned using pre-cleaning solvents such as acetone and isopropanol before loading into the chamber. We found the deposition parameters to be interdependent, with a relatively narrow processing window. Consequently, we optimized the TiN growth conditions by systematically varying the tooling factor, deposition rate, temperature, and pressure. An initial calibration was conducted by adjusting the tooling factor and deposition rate. The deposition was carried out at a rate of 1.5 Å s^{-1} with a tooling factor of 95. The chamber base pressure was maintained around 6×10^{-7} Torr before and after each deposition. The samples were annealed during deposition by adjusting the substrate heater temperature. We performed multiple depositions at varying substrate heater temperatures to monitor the film properties as a function of annealing temperature.

Optical properties were analysed using J.A. Woollam M-2000 spectroscopic ellipsometer in the spectral range between 193–1690 nm. The $\Psi(\lambda)$ and $\Delta(\lambda)$ parameters were measured at the following incident angles: 45°, 55°, 65° and 75°. The optical properties of TiN films were modelled using the Complete EASE software. All constituent layers of the investigated sample, i.e. the glass substrate, the TiN layer and the roughness were considered during modelling. The dielectric constants of glass were taken from the Woolam database and a General Oscillatory model was used to fit the dielectric constant of the deposited TiN films. Further, the thicknesses of the deposited samples were measured using ellipsometry and confirmed with a Profilm3D optical profilometer. The reflectance and transmittance spectra were recorded over the spectral range from 400 nm to 1000 nm, using a CCD spectrometer (USB-4000; Ocean Optics) and a white light-emitting diode .

The surface topography of the deposited TiN films was examined using an Atomic Force Microscope in the Scan Asyst mode over a Scan area of 2 μm $\times$ 2 μm. X-ray diffraction (XRD) measurements were performed with the Bruker D8 Discover diffractometer. A two-bounce Ge(022) monochromator and Eiger 2 R 500 K detector with 2D detector mode were used, along with 2.0 mm exit slit and 2.0 mm UBC collimator. All scans were performed in Bragg-Brentano geometry. The diffraction signal was integrated along the gamma axis across ROI (7.4° $\times$ 15.6°) to generate 2θ-intensity plots.

3. Results and discussion

Initially, the deposition parameter, i.e. annealing temperature, which significantly influences the microstructures and optical properties of the TiN films is analyzed. Figure 1(a) shows optical photographs of five 50 nm TiN films deposited under five different conditions: without heating, heating at 50 °C, 100 °C, 200 °C, and 300 °C respectively. It can be observed that the samples deposited at higher annealing temperatures are dark silver in colour, samples deposited for the temperatures 50 °C and 100 °C have golden luster, whereas the 'as deposited' sample is brownish in colour. This is because variations in annealing temperature significantly affect the stoichiometry of the films by altering the ratio of titanium/nitrogen atoms. At higher temperatures, nitrogen desorption from the TiN lattice creates nitrogen vacancies, even metallic Ti, under extreme conditions. Such a temperature-dependent stoichiometry variation directly affects the optical properties and hence the observed color of the films. As per previous studies, such temperature-dependent colour change is widely observed in refractory metal nitrides [25].

The optical properties calculated from the ellipsometry measurements are presented in figures 1(b) and (c) as the real ϵ_1 and imaginary ϵ_2 part of the complex dielectric function. From this figure we can see that the dielectric constant of TiN thin films can be tuned by varying the annealing temperature. The value of the real part of permittivity becomes most negative, (i.e. most plasmonic) with plasma frequency or ENZ wavelength λ_{ENZ} at 627 nm for the films deposited at 50 °C annealing temperature and the λ_{ENZ} point increases once the temperature increases. Unlike noble metals, changes in the material properties of TiN are often attributed to a reduction in carrier density rather than alterations in surface morphology [17]. Further, elevated temperatures can enhance grain growth and reduce defect density, which may improve film quality but also affect the plasma frequency by altering the effective electron density. Additionally, previous studies have suggested that surface oxidation of polycrystalline TiN at higher temperatures can lead to a shift in the plasma frequency. Interestingly, the sample annealed at 300 °C exhibited a double ENZ behaviour. This could be attributed to possible oxidation by residual oxygen [26], which produces titanium oxynitride (TiON) and titanium dioxide (TiO_2). This creates a heterogeneous structure in which metallic TiN coexists, with insulating or semiconducting phases, resulting in two separate ENZ wavelengths [27]. Some previous studies have reported a decrease in plasma frequency with increasing temperature, which aligns well with the decreasing trend observed in this study. Slight discrepancies may be attributed to variations in growth conditions, as previous studies primarily utilized sputtering techniques, with films deposited on a range of crystalline and nanocrystalline substrates. [21, 28].

In figures 2(a) and (b), the thickness-dependent permittivity of TiN films is shown for a constant annealing temperature of 50 °C, revealing a significant change in their metallic behaviour. It was observed that films with thicknesses up to 100 nm exhibit metallic characteristics, while those exceeding this thickness become dielectric in nature. Although the fine-tuning of film properties and deposition parameters is not valid over a broader temperature range, the optimized temperature window effectively serves the main aim of this study. In figure 2(c), a significant redshift in the λ_{ENZ} points is observed for the 50 nm TiN films annealed at 40 °C, 50 °C, and 60 °C. This demonstrates that even slight variations in annealing temperature can effectively tune the plasmonic properties from the visible to the NIR region. The real part of the permittivity value shows a monotonic increase with an increase in temperature, while the imaginary component shows a monotonic decrease (see figures 2(c) and (d)). Thin metal films ranging from 10 to 150 nm are ideal for plasmonic metamaterials and integrated optics and fall within the standard thickness range used in CMOS fabrication, allowing seamless microelectronic integration. This study focuses on sub-100 nm TiN films, particularly the highly plasmonic 50 nm films annealed at 50° C. Additionally, our low-temperature

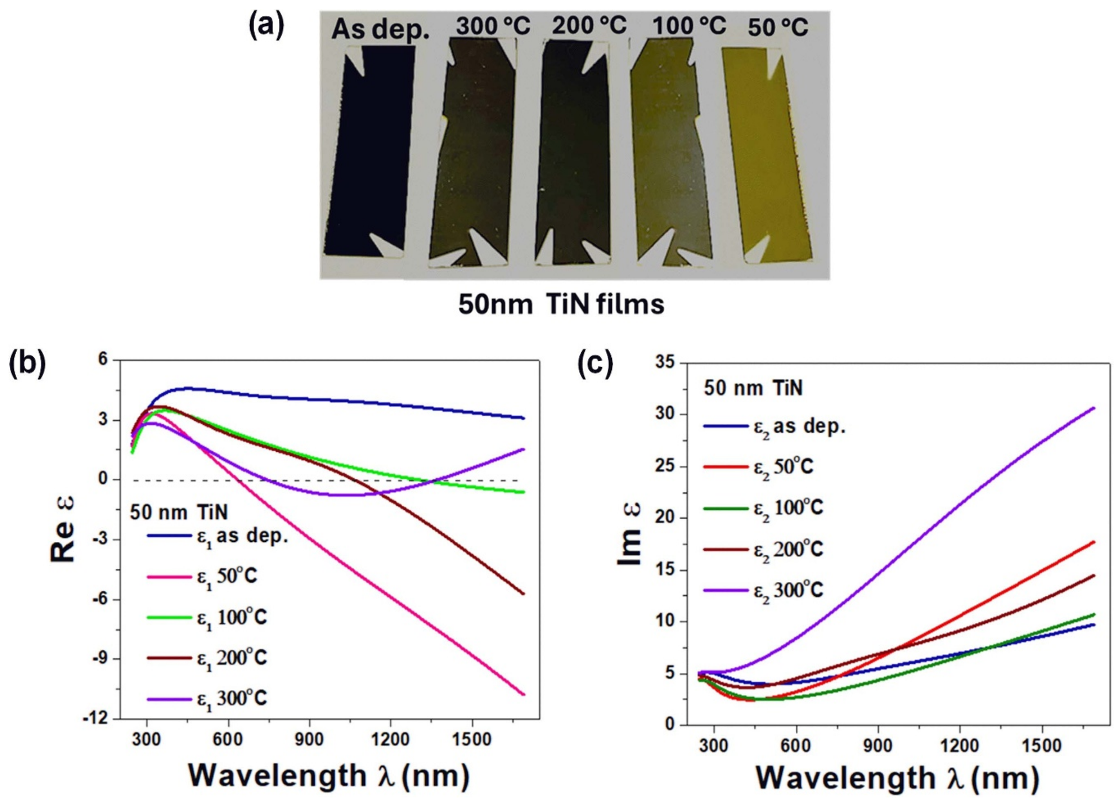

Figure 1. (a) Pictures of a series of five TiN samples, deposited with different substrate heating temperatures; Measured (b) real part and (c) imaginary part of the dielectric constant of TiN films: Change in permittivity for as-deposited samples and for samples annealed at 50 °C, 100 °C, 200 °C, and 300 °C, all with a thickness of 50 nm.

processing is compatible with flexible polymers and transparent substrates. As a result, further structural and morphological analyses were conducted on these films to gain deeper insight into their properties.

The 2D and 3D AFM micrograph of the 50 nm TiN film annealed at 50 °C is presented in figure 3(a). The observed granular pattern of the TiN film is characteristic of electron beam deposition, especially at low deposition temperatures. As a result, the optical effects of granularity, including surface roughness, are inherently incorporated into our analysis through ellipsometry measurements. Measuring surface roughness is essential, as it influences light scattering, film adhesion, electrical conductivity, and the overall efficacy of nanoscale plasmonic and photonic devices. A root-mean-square roughness value calculated over 2 μm $\times$ 2 μm surface area is 1.24 nm. The corresponding ellipsometry fitting gave an effective optical roughness of 2.35 $\pm$ 0.29 nm. The roughness values obtained from atomic AFM and ellipsometry is comparable with small discrepancies due to differences in measurement techniques. Further, the x-ray diffraction pattern of the same 50 nm TiN film deposited on a glass substrate is shown in figure 3(b). The sample exhibited face center cubic symmetry with peaks corresponding to (111), (200) and (220) planes located at $\sim$36.2 °, $\sim$42.5 ° and $\sim$61.8 ° respectively. The diffraction peak patterns observed in the samples were compared against the standard diffraction values provided in the JCPDS database (card no. 38-1420). The film is polycrystalline as indicated by the ring diffraction pattern in figure 3(b) bottom. The average crystallite size for the two orientations, (100) and (200), calculated using Scherrer's formula [29], is 13 nm. Nevertheless, the results show that the reported low-temperature e-beam evaporation process can be employed to fabricate high-quality plasmonic TiN films.

Figure 4 displays the unpolarized transmittance and reflectance spectra for 50 nm thick TiN films measured at normal incidence, where polarization effects in reflection and transmission become indistinguishable. The reflectance (R) reaches a minimum of $\sim$20% at λ $\sim$540 nm and rises sharply to $\geqslant$ 50% in the NIR region. Conversely, transmittance (T) peaks at $\sim$22% around λ $\sim$450 nm, then decreases with wavelength, falling below 5% in the NIR range. These trends closely mirror those of gold films, confirming the metallic nature of the TiN films [30]. The high plasmonic nature of the TiN films is further highlighted by their gold-like luster, as illustrated in the photograph inset in figure 3. Our e-beam evaporation method offers a simpler alternative to high-temperature sputtering [22] and ALD processes [18], while still maintaining a comparable ENZ response. The observed reflectance and transmittance trends closely match previously reported results [21, 31], demonstrating similar optical behavior for TiN thin films

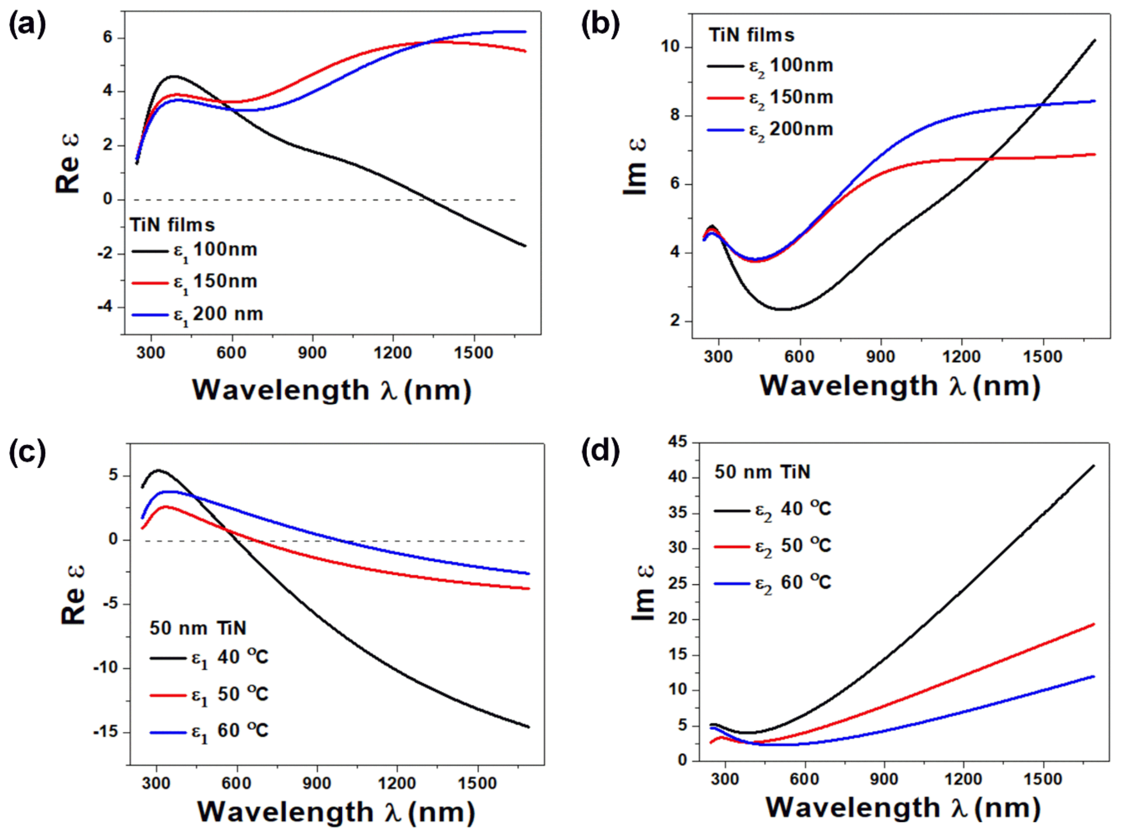

Figure 2. The change in (a) real part and, (b) imaginary part of the permittivity values for deposited samples with thicknesses of 100 nm, 150 nm, and 200 nm keeping the annealing temperature 50 °C; (c) and (d) wide tuning of the optical properties of metallic TiN films through slight variations in annealing temperature.

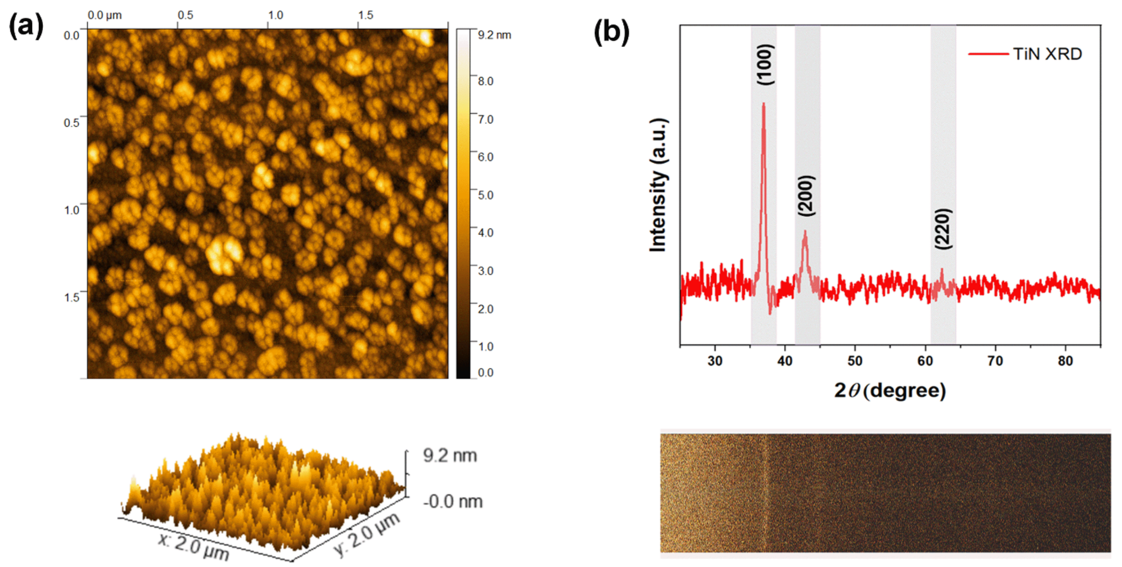

Figure 3. (a) 2D topography and the corresponding 3D AFM micrographs of the 50 nm TiN thin film on a glass substrate; (b) XRD spectra along with 2D XRD pattern of the corresponding deposited TiN film where the x-axis of the 2D XRD image corresponds to 2θ-axis.

in the visible to NIR range. More importantly, our low-temperature fabrication approach enables fine-tuning of optical properties by controlling deposition temperature and film thickness. This highlights e-beam evaporation as a simpler, more adaptable, and scalable deposition method.

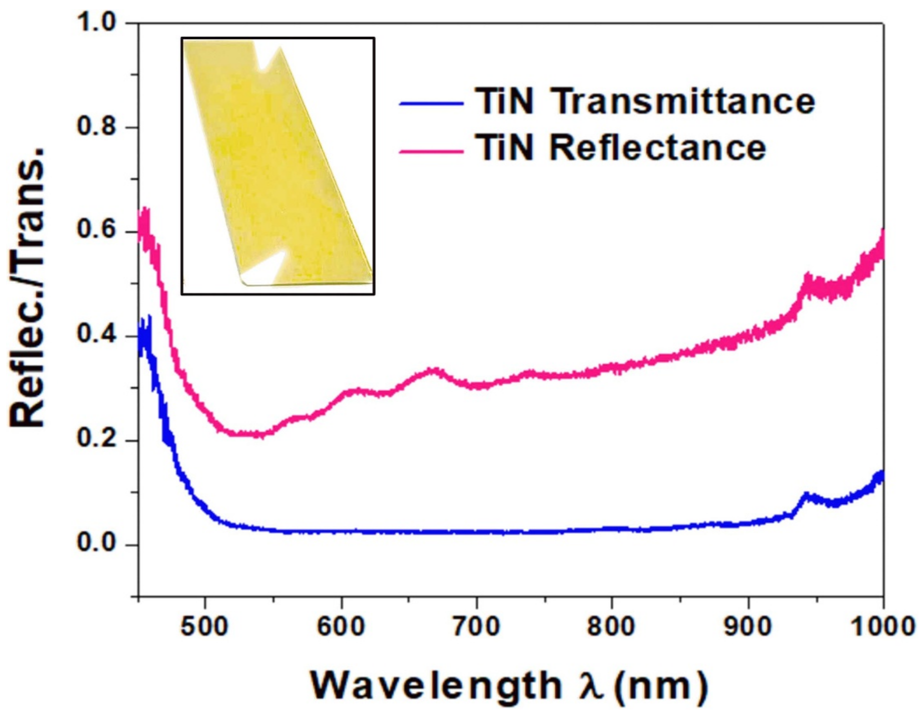

Figure 4. Transmittance and reflectance spectra of deposited TiN films on a glass substrate. The Inset image is a photograph of the TiN film.

4. Conclusions

An optimized strategy was proposed for depositing TiN thin films with enhanced plasmonic properties at low temperatures using electron beam evaporation. It was noted that while earlier studies primarily addressed thicker films, precise control of deposition parameters enables the adjustment of stoichiometry and microstructure in sub-100 nm films, significantly affecting their optical behaviour. The 50 nm TiN thin film annealed at 50° C, identified as having the best plasmonic properties, was subjected to detailed examination to analyze its crystallographic phase and optical behaviour. Additionally, the 50 nm TiN thin film annealed at 40° C demonstrated significantly larger negative ϵ_1 and high ϵ_2 values, making them particularly well-suited for certain loss-enabled plasmonic applications. For instance, in photon-assisted hot-carrier generation for photodetection and photocatalysis, a higher ϵ_2, enhances absorption and facilitates efficient carrier generation. Furthermore, plasmonic loss assisted solar and thermal energy harvesting, nanomanipulation, heat-assisted magnetic recording, and cancer photo-thermal therapy opening the door to the new field of thermoplasmonic application [32]. Further, our approach is compatible with a broader range of substrates, including flexible and thermally sensitive materials, making it simpler for use in new applications.TiN thin-film deposition at low temperatures has great promise for a range of photoplasmonic uses, such as surface plasmon-coupled emission , plasmonic solar cells, nanophotonic waveguides, plasmon-enhanced spectroscopy, and quantum optics [23, 33]. The trade-off between performance and fabrication flexibility makes our method particularly appealing for use in practical applications. This work provides a straightforward and cost-effective approach to fabricating TiN thin films, offering critical advancements for the development and optimization of TiN-based nanophotonic and plasmonic devices, particularly for practical high-power applications.

Data availability statement

The data that support the findings of this study are openly available at the following URL/DOI: https://doi.org/10.17630/42a1d567-9e02-4916-8709-e2b4911c715c.

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ORCID iDs

Yoshiko Nanao https://orcid.org/0000-0002-0376-2903
Akhil Rajan https://orcid.org/0000-0001-5356-3032
Sebastian A Schulz https://orcid.org/0000-0001-5169-0337

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